

<https://github.com/surabhi84ard/Mini/tree>

Abstract

We show how to use Pygmy as a decade counter.

1 Software

All codes used in this manual are available at the following link: <https://github.com/gadepall/vaman/tree/master/fpga/sevenseg/codes>

2 Setup

2.1 Pin Connections

Connect the pins in the bank J5 of the Pygmy with the seven segment display according to Table 1. Ensure that the COM pin is connected to 3.3V through a resistor.

Table 1: Seven segment display - Pygmy connection

Display	Pygmy
a	IO_4
b	IO_5
c	IO_6
d	IO_7
e	IO_8
f	IO_10
g	IO_11
COM	3.3V

2.2 Initial Code Execution

Flash the `helloworldfpga.bin` file to Pygmy. You should see the number 5 displayed.

2.3 Segment Assignments

The following lines are used for generating numbers on the display:

```
assign a = 0;
assign b = 1;
assign c = 0;
assign d = 0;
assign e = 1;
assign f = 0;
assign g = 0;
```

Modify the above code using Table 2 to generate numbers from 0–9.

Table 2: Pin values used for generating decimal numbers on the seven segment display

Decimal	abcdefg
0	1111110
1	0110000
2	1101101
3	1111001
4	0110011
5	1011011
6	1011111
7	1110000
8	1111111
9	1111011

3 Examples

3.1 Pin Configuration

Refer to Table 1 and the PU64 pin diagram in Figure 1 for pin numbering in the file `pygmy.pcf`.

3.2 7-bit Word Control

Execute the following code to display the number 4:

```
gpio_out = 7'b0100100;
```

3.3 Verilog Function for Decimal Input

Use a Verilog function to take a decimal input and display it on the seven segment display.
Code: `sevensseg_dec.v`

3.4 Decade Counter

Use the Pygmy as a decade counter. Code: `decade_counter.v`

4 Figures

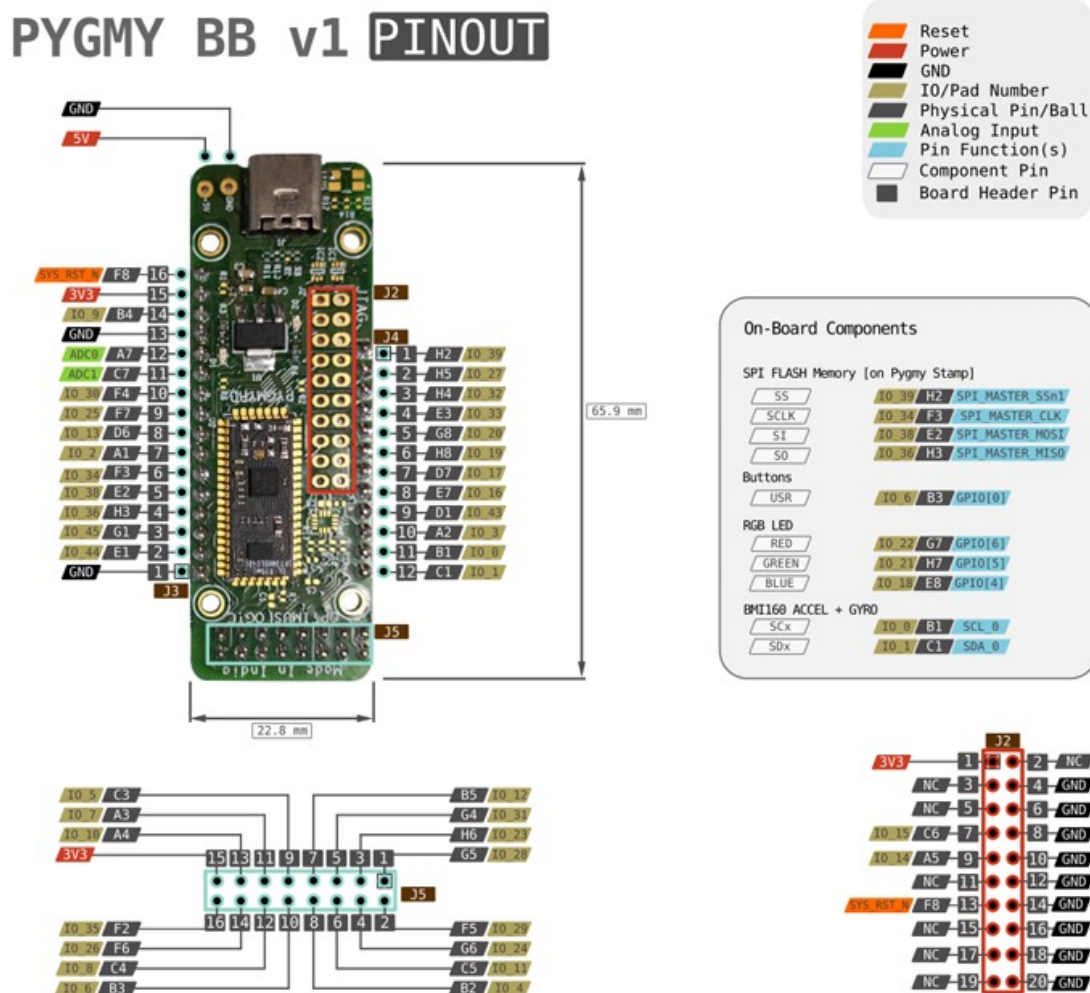


Figure 1: Pygmy BB v1 Pin Diagram